PRASUN KUMAR

AI/ML Technical Lead | GenAI Solution Architect | Agentic AI & RAG Specialist

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PROFESSIONAL SUMMARY

AI/ML Technical Lead and GenAI Solution Architect with 9+ years of software engineering experience leading the architecture and delivery of enterprise GenAI, RAG, Agentic AI, document intelligence, and AI testing solutions on Azure. Designs multi-tenant, production-ready platforms with observability, human-in-the-loop controls, idempotency, retries, privacy safeguards, and evaluation workflows. Improved document extraction accuracy by 60%+ through OCR normalization, prompt and schema design, confidence scoring, validation, and human review, while mentoring engineers and aligning technical delivery with stakeholder goals.

CORE EXPERTISE

Programming & Backend: Python, FastAPI, REST APIs, Microsoft Graph API, NumPy, pandas, PostgreSQL, MySQL

GenAI & LLM Engineering: Azure OpenAI, GPT, Llama, Gemma, Qwen, Mistral, IBM Granite, Prompt Engineering, Structured Extraction, Schema-Constrained Output

Agentic AI: LangGraph, LangChain, Multi-Agent Workflows, Tool and Function Calling, Typed State, Memory, Approval Gates, Human-in-the-Loop Controls

RAG & Search: Azure AI Search, Embeddings, Vector Search, Semantic and Hybrid Retrieval, Chunking, Metadata Filtering, Citations, Retrieval Evaluation

Machine Learning & NLP: scikit-learn, XGBoost, BERT, RoBERTa, Sentence Transformers, PyTorch, TensorFlow, Hugging Face, Classification, Recommendation Systems

Azure & Integration: Azure Functions, Azure Service Bus, Azure Blob Storage, Azure AI Document Intelligence, Event-Driven Architecture, Webhooks, Case-Management Integrations

MLOps, LLMOps & Platform: Databricks, MLflow, Docker, Kubernetes, CI/CD, Model and Prompt Versioning, Evaluation Gates, Monitoring, Lifecycle Governance

Architecture & Leadership: Solution Architecture, Multi-Tenant SaaS, Idempotency, Retries, DLQs, Observability, Privacy Controls, Design and Code Reviews, Mentoring, Stakeholder Alignment

PROFESSIONAL EXPERIENCE

Associate AI Specialist (G3A) | Harbinger Group | Jul 2026 - Present

Associate Tech Specialist (G3A) | Harbinger Group | Jul 2024 - Jun 2026

Across lead-level AI/ML roles since Jul 2024, lead architecture and delivery for enterprise GenAI, Agentic AI, RAG, document intelligence, AI automation, and AI evaluation initiatives across Azure-based workflows.

Architected HAS multi-tenant SaaS capabilities spanning Discovery Agent, Automation Tester, and EvalAI Pro using Azure Functions, Azure Service Bus, Azure Blob Storage, PostgreSQL, Azure OpenAI, Azure AI Search, and Jira/Azure DevOps integrations.

Architected a GenAI email-to-case platform using Microsoft Graph API, Azure Functions, Azure Service Bus, Azure Blob Storage, Azure OpenAI, OCR/Azure AI Document Intelligence, MySQL, case-management APIs, idempotency, retries, DLQs, audit trails, and human review.

Improved document-extraction accuracy by 60%+ through OCR normalization, prompt and schema design, confidence scoring, validation, and human-review loops.

Designed enterprise RAG patterns covering ingestion, metadata-aware chunking, embeddings, semantic and hybrid retrieval, Azure AI Search, citations, confidence thresholds, and retrieval evaluation.

Led behavior-scoring and text-classification work using BERT, RoBERTa, Sentence Transformers, and XGBoost patterns plus Llama, Gemma, and Qwen experimentation, threshold calibration, and Databricks evaluation dashboards.

Reduced model and storage footprint by more than 77 TB through lifecycle cleanup, governance, artifact review, and storage optimization.

Mentor engineers; lead architecture, design, and code reviews; manage stakeholder discussions; and define production-readiness checks.

Senior Software Engineer | Harbinger Group | Apr 2021 - Jun 2024

Delivered enterprise software and AI/ML-adjacent systems across backend services, automation workflows, integrations, document processing, and data-driven applications.

Built and integrated APIs, data-processing flows, automation utilities, and reporting components with emphasis on maintainability and reliability.

Collaborated with product, QA, DevOps, and client stakeholders to translate business requirements into scalable technical implementations.

Improved engineering delivery through code reviews, reusable patterns, troubleshooting support, and mentoring of junior engineers.

Software Professional | Extentia Information Technology Pvt. Ltd. | Nov 2019 - Apr 2021

Developed backend, ETL, and integration solutions using Python, pandas, REST APIs, database workflows, and cloud-aligned delivery patterns.

Implemented NLP chatbot and AI automation prototypes spanning intent recognition, entity extraction, BERT-style embeddings, Elasticsearch, resume parsing and ATS matching, translation API patterns, and computer-vision attendance workflows.

Associate IT Applications Specialist | Symantec Software India Pvt. Ltd. | Jun 2017 - Oct 2019

Supported enterprise application engineering, automation, data workflows, and operational reliability for internal business systems.

Built Python automation for database patching, refreshes, MySQL backups, SOX audit reporting, and LMS JSON-template conversion.

Created Flask/MySQL dashboards for ServiceNow tickets and server-performance visibility, improving operational productivity by 30%.

Master Level Intern | Symantec Software India Pvt. Ltd. | Jan 2017 - Jun 2017

Worked on enterprise application support and automation foundations before moving into full-time engineering.

SELECTED ARCHITECTURE PROJECTS

GenAI-Powered Email-to-Case Automation Platform (Technical Lead / AI Solution Architect): Microsoft Graph API ingestion, Azure Blob Storage, Azure Service Bus, Azure Functions, OCR/Azure AI Document Intelligence, Azure OpenAI extraction, MySQL, validation, confidence scoring, idempotency, retries, DLQ handling, audit trails, and human review.

Enterprise RAG Knowledge Platform (RAG Architect): Document ingestion, metadata-aware chunking, embeddings, Azure AI Search vector and hybrid retrieval, retrieval APIs, citations, confidence thresholds, quality metrics, and hallucination-risk reduction.

AI-Driven Automation Tester Platform (AI Automation Architect): Jira/Azure DevOps work-item intake, LLM-generated test cases, Playwright API/UI script generation, Azure Service Bus handoffs, Blob report storage, Xray/ADO result synchronization, and defect routing.

Agentic AI Multi-Agent Platform (Agentic AI Specialist): Planner, research, retrieval, validation, execution, dynamic tool routing, memory, human approval, retries, checkpoints, and audit-trace patterns using LangGraph-style orchestration.

AI Behavioral Scoring Platform (Technical Lead): NLP and text-classification experimentation using BERT, RoBERTa, Sentence Transformers, XGBoost, Llama, Gemma, and Qwen patterns, with multilingual considerations, threshold calibration, and evaluation dashboards.

EvalAI Pro and LLM Evaluation (LLMOps / Evaluation): Agent and RAG evaluation patterns using RAGAS, DeepEval, MLflow-style tracking, run metrics, drift analysis, regression gates, and artifact persistence for release-readiness decisions.

SELECTED TECHNICAL PORTFOLIO

enterprise-ai-ml-case-studies: Sanitized case studies and runnable POCs for GenAI, RAG, document AI, agentic workflows, NLP, and computer vision, with tests, Docker, CI, architecture, evaluation, and production-readiness notes.

agentic-ai-langgraph-workflows / enterprise-rag-quality-lab: Typed-state agent workflows with approval, retry, tool, and audit patterns plus retrieval, grounding, citation-coverage, and confidence-based review evaluation.

document-intelligence-llm-pipeline / email-to-case-genai-automation: Privacy-aware extraction, schema and confidence gates, idempotency, review routing, validation, and audit-trail reference implementations.

production-llm-platform-on-kubernetes / privacy-preserving-rag-dlp: Portfolio reference implementations for a FastAPI LLM gateway, policy and usage checks, metrics, Kubernetes manifests, privacy scanning, redaction, and citation-aware retrieval.

EDUCATION & CREDENTIALS

Executive Post Graduate Programme in Machine Learning and Artificial Intelligence, Specialization: Advanced Deep Learning | IIIT Bangalore | 2023 | CGPA: 2.92/4.00 | Credential: credential.net/20453200-981b-4985-9ddf-2a44c5ee66cf

Master of Computer Applications (MCA) | Savitribai Phule Pune University | 2017 | 69.28% | CGPA: 4.110 | Grade B

Bachelor of Computer Applications (BCA) | Sikkim Manipal University | 2014 | 65.78% | Grade B

AWARDS & RECOGNITION

3x Quarterly Team Award Winner; 2x Superstar Award Winner; 4x Technical Star Recognition.

Additional recognition includes We-NOT-Me, Going the Extra Mile, Thank You, FunStar, PEP Silver, PEP Gold, and Hackathon Recognition (2017), spanning technical excellence, innovation, leadership, ownership, collaboration, and delivery impact.